

Math 241, F12
Quiz 8

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (3 points) Use Chain Rule to find $\partial z/\partial s$ and $\partial z/\partial t$ when

$$z = e^{x+2y}, \quad x = s/t, \quad y = t/s$$

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = e^{x+2y} \left(\frac{1}{t} \right) + 2e^{x+2y} \left(-\frac{t}{s^2} \right) \\ &= e^{x+2y} \left(\frac{1}{t} - \frac{2t}{s^2} \right) = e^{\left(\frac{s}{t} + \frac{2t}{s} \right)} \left(\frac{1}{t} - \frac{2t}{s^2} \right) \\ \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = e^{x+2y} \left(-\frac{s}{t^2} \right) + 2e^{x+2y} \left(\frac{1}{s} \right) \\ &= e^{x+2y} \left(-\frac{s}{t^2} + \frac{2}{s} \right) = e^{\left(\frac{s}{t} + \frac{2t}{s} \right)} \left(-\frac{s}{t^2} + \frac{2}{s} \right) \end{aligned}$$

Problem 2. (3 points) Find the directional derivative of the function $f(x, y) = \sin(xy)$ at the point $(1, 0)$ along the direction of the vector $\mathbf{v} = \langle 1, 2 \rangle$. What is the maximum rate of change of f at this point?

$$\begin{aligned} \nabla f(x, y) &= \langle f_x(x, y), f_y(x, y) \rangle \\ &= \langle y \cos(xy), x \cos(xy) \rangle \end{aligned}$$

$$\Rightarrow \nabla f(1, 0) = \langle 0, 1 \rangle$$

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\langle 1, 2 \rangle}{\sqrt{1^2 + 2^2}} = \frac{1}{\sqrt{5}} \langle 1, 2 \rangle = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$

$$\Rightarrow D_{\mathbf{u}} f(1, 0) = \nabla f(1, 0) \cdot \mathbf{u} = \langle 0, 1 \rangle \cdot \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle = \frac{2}{\sqrt{5}}$$

The maximum rate of change of f at $(1, 0)$ is

$$|\nabla f(1, 0)| = \sqrt{0^2 + 1^2} = 1.$$

Problem 3. (4 points) Find the local maximum and minimum values and saddle points of the function

$$f(x, y) = x^4 + y^4 - 4xy + 2.$$

$$f_x(x, y) = 4x^3 - 4y \quad f_y(x, y) = 4y^3 - 4x$$

We solve for critical points

$$\begin{cases} 4x^3 - 4y = 0 \\ 4y^3 - 4x = 0 \end{cases} \Rightarrow \begin{cases} x^3 - y = 0 \\ y^3 - x = 0 \end{cases} \Rightarrow y = x^3$$

$$\Rightarrow (x^3)^3 - x = 0$$

$$x^9 - x = 0$$

$$x(x^8 - 1) = 0$$

$$x(x-1)(x+1)(x^2+1)(x^4+1) = 0$$

$$\Rightarrow x = 0, x = 1, x = -1$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ y = 0 & y = 1 & y = -1 \end{matrix}$$

$\Rightarrow f$ has three critical points $(0, 0)$, $(1, 1)$, $(-1, -1)$

$$f_{xx}(x, y) = 12x^2, \quad f_{yy}(x, y) = 12y^2, \quad f_{xy} = -4.$$

For $(0, 0)$, we have

$$D(0, 0) = f_{xx}(0, 0)f_{yy}(0, 0) - f_{xy}^2(0, 0) = -16 < 0$$

$\Rightarrow (0, 0)$ is a saddle point.

For $(1, 1)$, we have

$$D(1, 1) = f_{xx}(1, 1)f_{yy}(1, 1) - f_{xy}^2(1, 1) = 12 \cdot 12 - 16 = 128 > 0$$

$$f_{xx}(1, 1) = 12 > 0$$

$\Rightarrow f(1, 1) = 1 + 1 - 4 + 2 = 0$ is a local minimum value.

For $(-1, -1)$, we have

$$D(-1, -1) = f_{xx}(-1, -1)f_{yy}(-1, -1) - f_{xy}^2(-1, -1) = 128 > 0$$

$$f_{xx}(-1, -1) = 12 > 0$$

$\Rightarrow f(-1, -1) = 0$ is also a local minimum value.